Designing a JK Serial adder

- $J_Q = x \cdot y$
- $K_Q = x' \cdot y' = (x+y)'$
- $S = x \oplus y \oplus Q$



Present State (Q)	inputs x	y	next State (Q)	output (S)
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

Counters

Def: register that goes through a prescribed series of states

types:
- asynchronous (ripple) → flip-flop output serves as a clock for triggering connected flip-flops
- synchronous → all flip-flops are triggered by a clock signal at the same time.

Counter design procedure

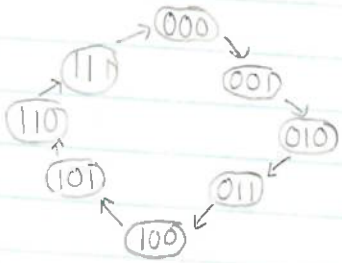
1 - Draw state table & diagram of output sequence
2 - Determine F-F type & make transition table
3 - Build state table with F-F inputs indicated
4 - Use K-maps to record & simplify the available outputs to create inputs.
5 - Implement the circuit.

Binary ripple counter

- Circuit that counts clock pulses & displays the running total in binary
- each F-F controls the CLK input of the next F-F
- F-F do not change states in exact synchronism with the applied clock pulse
- rippling effect (kinda like a domino)

A_3	A_2	A_1	A_0
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0

* reset signal sets all outputs to 0.



* number of output states is called the modulus or mod.

full sequence count: natural count that includes all possible binary numbers
modulus = max modulus

Truncated sequence count: modulus < max modulus less than all possible binary numbers are used



state diagram

Circuit			next		
Q_c	Q_b	Q_a	Q_c	Q_b	Q_a
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	0	0	0

state table